

P1 — Remote Sensing + AI/ML

SIH 26143 · Detailed 6-Phase Work Plan · 6 owned work packages

ROLE OBJECTIVE

Own the SAR/ML detection layer and make model outputs measurable, georeferenceable and usable by reconstruction and attribution.

HOW TO USE THIS PLAN

Each package has four parts: what to build/do, the concrete deliverable, dependencies, and an acceptance condition. The owner is responsible for making the output usable by the next phase—not merely completing a task locally.

PHASE 1 — DATA & PREPROCESSING

Work package 1: Sentinel-1 preprocessing pipeline

Set up the SAR ingestion structure; inspect VV/VH channels; perform radiometric preparation, normalization and appropriate speckle handling; preserve scene metadata and create reproducible preprocessing configurations; generate model-ready tiles without breaking image/mask correspondence.

Deliverable: reproducible preprocessed SAR inputs + preprocessing configuration + sample before/after checks.

Dependencies / acceptance: Depends on: dataset audit by P6. Acceptance: same input/config produces the same model-ready output and no image/mask pairing is lost.

PHASE 2 — SPILL DETECTION

Work package 2: U-Net segmentation model

Build the baseline U-Net in PyTorch; define input channels, loss function, train/validation loop, augmentation policy and checkpointing; train on the primary binary-mask dataset; save inference weights and configuration.

Deliverable: trained U-Net checkpoint + inference script + training configuration.

Dependencies / acceptance: Depends on: P1 preprocessing and P6 split verification. Acceptance: inference produces a binary/soft spill mask for an unseen scene.

PHASE 3 — SOURCE RECONSTRUCTION

Work package 3: Spill morphology features

Convert segmentation output into measurable geometry: connected components, area, centroid, bounding geometry, orientation and shape descriptors; distinguish candidate regions and retain confidence information for downstream processing.

Deliverable: structured spill-feature object containing geometry, area and model confidence.

Dependencies / acceptance: Depends on: U-Net output and geospatial reference from P4/P2. Acceptance: dashboard/backend can consume the same geometry and metadata.

PHASE 4 — AIS CORRELATION

Work package 4: Detection-to-AIS feature interface

Define the exact hand-off from detection/reconstruction to AIS analysis: incident ID, detection time, spill geometry, reconstructed source corridor, origin-time window and uncertainty fields; document units and coordinate systems.

Deliverable: versioned feature contract/schema for the attribution layer.

Dependencies / acceptance: Depends on: P2 API schema and P5 reconstruction outputs. Acceptance: P4 can run correlation without manually reformatting ML output.

PHASE 5 — EVIDENCE FUSION + UI

Work package 5: Detection confidence integration

Expose segmentation confidence and look-alike indicators as evidence inputs; separate model confidence from physical/source confidence; ensure low-quality or ambiguous detections can be flagged rather than silently treated as confirmed oil.

Deliverable: detection-evidence payload + uncertainty/quality flags.

Dependencies / acceptance: Depends on: P6 look-alike evaluation. Acceptance: ambiguous detection can propagate an explicit warning downstream.

PHASE 6 — VALIDATION & DEPLOYMENT

Work package 6: Final ML evaluation

Run held-out evaluation; calculate IoU, Dice, precision and recall; analyse false positives on look-alikes; record failure examples by scene condition; compare against the agreed baseline only after evaluation is complete.

Deliverable: ML evaluation report, error gallery and final model card.

Dependencies / acceptance: Depends on: complete dataset split and integrated pipeline. Acceptance: every reported metric has a defined split and reproducible evaluation script.

CROSS-TEAM HANDOFF RULE

No component is considered complete if its output cannot be consumed by the next owner. Preserve IDs, timestamps, coordinate systems, provenance, uncertainty and failure states across every handoff.

Core project rule: AIS gaps, uncertain metocean forcing, ambiguous SAR signatures or weak evidence must increase uncertainty or trigger abstention—not be converted into fabricated confidence.